Mathematics & Statistics Faculty

NAME: ____________________

Teacher: DRD GIB HPA/GIB WNA TRR

Level 2 Mathematics & Statistics

AS 91261 Apply algebraic methods in solving problems

CREDITS: Five

ACHIEVEMENT	ACHIEVEMENT WITH MERIT	ACHIEVEMENT WITH EXCELLENCE
Apply algebraic methods in solving problems.	Apply algebraic methods, using relational thinking, in solving problems	Apply algebraic methods, using extended abstract thinking, in solving problems.

You should answer ALL parts of ALL questions in this booklet.

You should show ALL your working.

Make sure that you have the Formulae sheet

If you need more space for any answer, use the space provided at the back of the booklet then ask the exam supervisor if you need more.
Clearly number all extra working.

Q1	Q2	Q3	**TOTAL**	**GRADE**

QUESTION ONE

(a) Simplify the expression, leaving your answer with positive indices $\left(\frac{2x^3y^2}{x^2}\right)^{-2}$

(b) If $(2-3x)\times f = 15x^2 - 22x + 8$

Find f in terms of x and express it in its simplest form.

(c) If $x = log2$ and $y = log5$, write $log\sqrt[3]{10}$ in terms of x and y.

(d) Make x the subject of the formula $p = \sqrt{\frac{2-ax^2}{x^2-b}}$

(e) The following quadratic equation, where m is a constant, has two distinct real roots.

$$x^2 + (m+2)x + 4m - 7 = 0$$

Determine the range of possible values for m.

(f) Given that $a = x^{\frac{1}{2}} + x^{-\frac{1}{2}}$ and $b = x^{\frac{1}{2}} - x^{-\frac{1}{2}}$

Show that $a^2b^2 + 4 = \left(x + \frac{1}{x}\right)^2$

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QUESTION TWO

(a) Write as a single logarithm in simplest form: $log x^3 + log\frac{1}{x} - 5log x$

(b) A function is defined as $f(x) = x^2 - 10x + 4$

Express $f(x)$ in completed square form, i.e. $f(x) = (x + a)^2 + b$, where a and b are integers.

(c) A car purchased new for $52,000 in January 2014 depreciates at a rate of 20% per year.

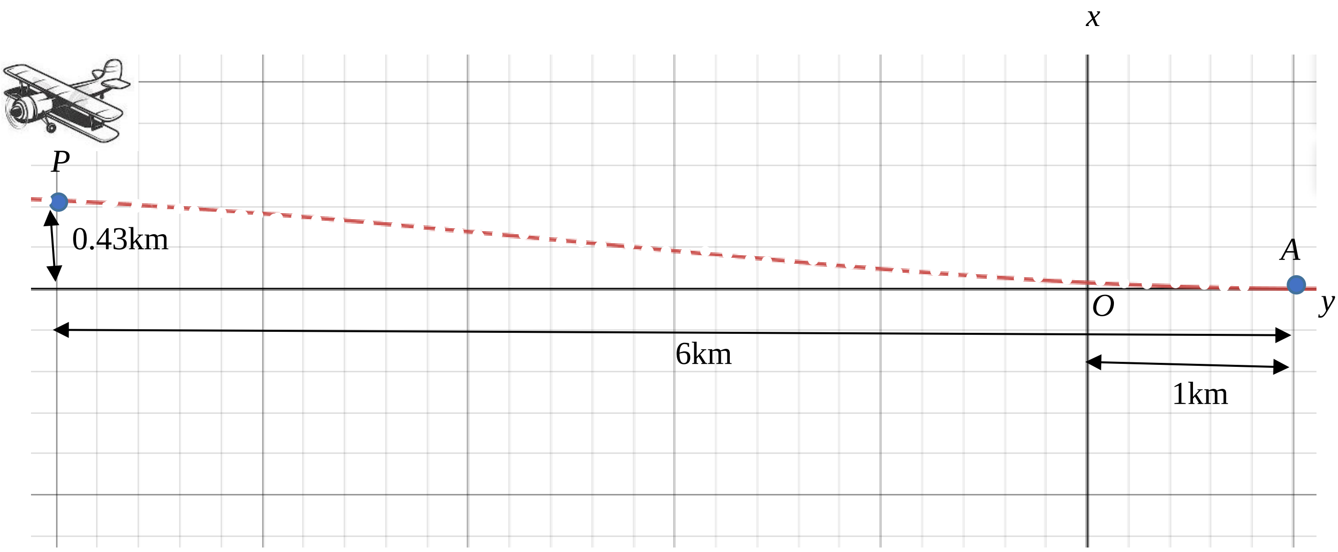
Its value V after t years, in thousands of dollars, can be modelled by: $V = 52 \times 0.8^t$.

(i) Use algebra to find after how many years was the car worth $24,000?

A second car, initially worth $39,500 in January 2014 loses value at 15% per year.

(ii) In what year were the two cars worth the same? Show all algebraic working.

(d) A plane is flying at a height of 0.43km above ground level when it begins its descent 6km from the airport at point A. The airport is 1km from the origin O.



Taking O as the origin, the curve (shown above as a dotted line) approximately describes the path of the plane's descent where $f(x) = 0.0024x^3 + 0.018x^2 - cx + d$.
Use algebra to find the value of c and d and hence state the full function which approximates the plane's descent.

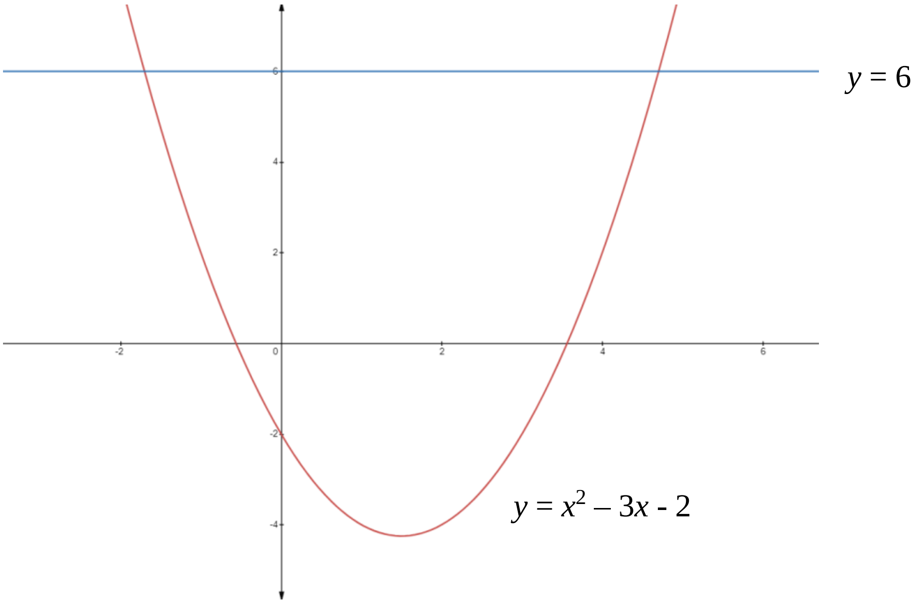
QUESTION THREE

(a) Simplify $\frac{x}{x+5} + \frac{2}{x-3}$

(b) Solve. $\frac{5x^2-72}{x-3} = 2x$

(c) Find the nature of the roots of $7a^2 + 5(a+1) = 3$

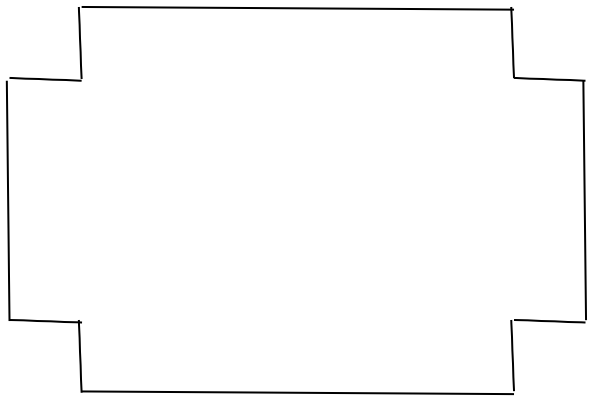
(d) Use algebra to find the **exact** x values where the parabola $\boldsymbol{y = x^2 - 3x - 2}$ crosses the line $\boldsymbol{y = 6}$ as shown below.



(e) A rectangular piece of sheet metal is 7 cm shorter than it is wide.

From each corner, a 3×3 cm square is cut out and the flaps are then folded up to form an open box.

If the volume of the box is 1620 cm^3, find the length and width of the original piece of sheet metal.



(f) The quadratic equation $3(k+2)x^2 - (5k+7)x + 3k + 1 = 0$
where k is a constant and $k \neq -2$, has two distinct real roots.

Use algebra to justify that $\frac{-25}{11} < k < 1$

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